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DETAILED SPECIFICATIONS
ARPEX Project

[YEAR 2 OF THE PROJECT]

This document covers only the work of the second year

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1 TECHNICAL SPECIFICATIONS – CHASSIS

1.1 Phase 1 – Redesign of the chassis and front suspension

The chassis of the 2024–25 concept car has shortcomings that will be documented in the 2025–2026 report. The existing chassis is therefore not reused as it is: an almost complete redesign was needed, especially around the suspension mounting points (pushrod, wishbones).

The aim of this phase is to start from the existing model in order to:

- analyse precisely what does not work (geometry, local stiffness);
- understand how these defects affect safety (risk of failure, loss of control) and the structural and aerodynamic behaviour of the vehicle;
- review and use the documentation already produced (PLP reports, SolidWorks/ANSYS models, simulation results);
- propose a new tubular front architecture inspired by solutions used on hypercars and in Formula Student (better load distribution).

The new chassis must remain compatible with the fixed overall dimensions (concept-car envelope), while integrating the new constraints related to the suspension, the electric motor, the batteries and the future bodywork.

1.2 Phase 2 – Modelling and preparing the calculations

Once the new architecture is defined, the chassis will be fully remodelled in SolidWorks, starting from the existing tubular model. The steel chosen for the structure remains an AISI 4130-type steel, a standard material for competition tubular chassis (good compromise between strength, weldability and mass).

This model aims to:

- correctly integrate the new suspension rod mounting (pushrod moved from the lower to the upper wishbone);
- check the load paths (how forces travel through the tubes);
- prepare a clean model for finite element analysis: properly defined tubes, consistent sections, geometric continuity.

The calculation model will then be exported to ANSYS Mechanical, using beam elements (BEAM model) suited to tubular structures, with a mesh fine enough to capture the critical zones without making the calculations unusable.

1.3 Phase 3 – FEA: test protocol

The finite element analysis (FEA) aims to verify that the modified chassis remains strong and stiff enough in every realistic scenario for a track vehicle. Rather than relying on a single load case, a series of numerical tests will be run to cover extreme situations.

The calculations will be carried out in ANSYS Mechanical using the following protocol:

1. Top-speed test

This test represents the vehicle running in a straight line at high speed, with its full weight and the aerodynamic downforce applied to the structure. An overall vertical force is applied at the centre of gravity, while the front is fixed and the rear remains free to move along the longitudinal axis. **Objective:** check that the chassis can carry the maximum vertical loads without difficulty (no tube close to failure, small and reversible deformations).

2. Tight-corner test

Here we simulate a strong lateral acceleration in a corner, as when heavily loading the car on corner entry. A lateral force is applied at the centre of gravity, distributed through several rigid links.
Objective: make sure the chassis does not deform excessively sideways and that the areas around the front suspension (wishbones, pushrod) remain at acceptable stress levels.

3. Maximum-braking test

This case corresponds to emergency braking at high speed. Longitudinal braking forces, combined with a vertical load, are applied to the structure.
Objective: check that the front mounting points, the links to the floor and the areas near the front axle hold without crack initiation or local buckling.

4. Maximum-acceleration test

This case mirrors braking, but with maximum traction from the powertrain. A rearward longitudinal force, combined with the vertical load, is applied to the rear of the chassis.
Objective: validate the strength of the rear area (motor mounts, rear suspension mounting points) and make sure the structure does not deform permanently.

5. Torsional stiffness test (driving over an obstacle)

Finally, a global torsion test represents driving over an obstacle (bump, kerb) that lifts only part of the vehicle. A torsional moment is applied between the front and the rear through rigid links.
Objective: measure the torsional stiffness of the chassis (rotation angle for a given torque) and check that it is in line with expectations for a track vehicle (a chassis "firm" enough for good driving feel).

1.4 Summary of stiffness criteria

With a vehicle + driver mass of 830 kg (8142 N) and a chosen steel yield strength of 460 MPa, the results obtained are given below. For each case, stresses are evaluated tube by tube at two levels of analysis: overall maximum stress and local stress (centre of the tube).

Case 1 – Top speed	
Overall max stress	661 MPa
Max stress at tube centres	147 MPa
Overall max strain	0.3 %
Max strain at tube centres	0.1 %

Table 1: Expected values – top-speed case

Case 2 – Tight corner	
Overall max stress	206 MPa
Max stress at tube centres	45.8 MPa
Overall max strain	0.098 %
Max strain at tube centres	0.022 %

Table 2: Expected values – tight-corner case

Case 3 – Maximum acceleration/braking	
Overall max stress	362 MPa
Max stress at tube centres	121 MPa
Overall max strain	0.17 %
Max strain at tube level	0.0574 %

Table 3: Expected values – maximum acceleration/braking case

Case 4 – Torsion / obstacle	
Applied moment	5280 N·m
Max rotation	0.016839 rad $\approx 1°$

Table 4: Expected values – torsion / obstacle case

1.5 Phase 4 – Critical points to monitor after modification

Moving the pushrod mounting from the lower wishbone to the upper wishbone changes the load path inside the chassis. The following areas will therefore be studied with particular care:

- the tube joints around the suspension mounting points;
- the continuity of the main load-bearing tubes, to avoid creating "hard points" or stress concentrations;
- the impact on the overall suspension geometry (heights, angles, travel), in line with the handling objectives.

1.6 Battery integration

The chassis must be designed to hold the vehicle's main battery pack in **two possible configurations**, without major changes to the structure:

- "short range" configuration: **5 batteries**,
- "long range" configuration: **9 batteries**.

Each battery unit has the following characteristics:

- Outer dimensions: 59 cm (L) × 22 cm (W) × 11 cm (H),
- Unit volume: 14.3 L,
- Unit mass: 32 kg,
- Nominal voltage: 30 V,
- Capacity: 234 Ah,
- Nominal energy: 6.85 kWh,
- Usable energy: 4.11 kWh (20–80 % state-of-charge window).

The two configurations therefore represent total masses and energies of about:

- **5 batteries:** 160 kg, about 34 kWh nominal (≈21 kWh usable),
- **9 batteries:** 288 kg, about 62 kWh nominal (≈37 kWh usable).

Chassis requirements:

- Provide enough space to house 9 batteries of the above dimensions, with mounting solutions that allow only 5 to be fitted without changing the structure.
- Allow a careful mass distribution: the batteries must be placed as low as possible and as close as possible to the centre of gravity to limit pitch and roll.
- Guarantee access for assembly and maintenance (removing one battery without taking out the whole pack).
- Integrate high-voltage cable routing and any cooling interfaces (air or liquid) from the design stage.

1.7 Phase 5 – Final validation

The final validation of the modified chassis will be declared only if all the tests described above are passed. In practice, as long as these conditions are not met simultaneously, the chassis geometry will be considered as still being iterated and must be adjusted (reinforcements, repositioned tubes, local changes) before a new round of calculations.

2 TECHNICAL SPECIFICATIONS – AERODYNAMICS

2.1 Context and objectives

Aerodynamics is critical for a high-performance vehicle running at 200 km/h on track. Our objective is to optimise the bodywork and flat floor to minimise drag (Cx) while maximising downforce (Cz) and ensuring directional stability.

Aerodynamic specifications

Aerodynamic parameters	Target	Unit
Drag coefficient (Cx)	0.4–0.6	—
Overall downforce coefficient (Cz)	1.6–2.4	—
Downforce/drag ratio	4:1	—
Aerodynamic balance	50/50 ±5 %	—
Top speed	200	km/h

Table 5: Aerodynamic specifications

2.2 Objective 2: Aerodynamic study of the bodywork and flat floor

Phase 1 – Research and documentation
To design the optimal bodywork, we analysed existing solutions and studied the best-performing technologies:

- **Bugatti Bolide:** hypercar that inspires our concept (optimised front/rear geometry)
- **Formula SAE:** student competition cars with advanced aerodynamic optimisation (wings, diffusers)
- **Sporty electric vehicles:** Porsche Taycan, Tesla Roadster (airflow management, battery cooling)
- **Modern flat floor:** technique borrowed from current Formula 1, essential to create low pressure under the vehicle

This phase lets us identify best practices and adapt these technologies to our electric racing concept car.

Specific objectives:

1. Reach Cx $\leq$ 0.60
2. Generate net downforce $\geq$ 4 times the vehicle drag at 200 km/h
3. Optimise battery cooling (integrated air intake)
4. Ensure stability (no pitch-up at high speed)

Phase 2 – 3D bodywork modelling
In Fusion 360, creation of a body made of 3 models:



Figure 1: McLaren Solus GT (front of our car)



Figure 2: Ferrari Vision (middle of our car)



Figure 3: Bugatti Bolide (rear of our car)

Phase 3 – Chassis compatibility check

Before finalising the bodywork, we will check that it fits perfectly with all the other parts of the vehicle, with no conflict.

We will virtually assemble all critical components in the 3D software:

- **Corrected chassis:** check that the bodywork mounting points match exactly
- **Electric motor and inverter:** make sure the motor can move freely (full travel) without touching the bodywork
- **Battery:** exact positioning with space for cooling and wiring
- **Bucket seat:** check that it fits perfectly in the cockpit with easy access
- **Driver:** model an average-sized driver (1.70 m, 70 kg) in driving position to confirm interior space and visibility

This step ensures that the aerodynamically optimised body remains functional and usable in the complete vehicle.

Phase 4 – CFD simulations (STAR-CCM+)

To analyse how air behaves around the vehicle, we will use STAR-CCM+. The computational domain will deliberately be quite large (10 times the vehicle length) to capture all the important aerodynamic phenomena: the vortices created by the vehicle, the recirculation zones at the rear, and the interaction with the ground.

We will pay particular attention to the air layers close to the body (near-wall region), since this is where drag and downforce are generated. We will use a very fine mesh with prism-layer refinement (20 layers) to obtain reliable and accurate results.

This approach will let us optimise the vehicle shape to reduce drag while increasing downforce.

Simulation conditions:

Parameter	Value	Justification
Relative velocity	55.5 m/s (200 km/h)	Close to track top speed
Turbulence regime	—	Turbulent regime confirmed
Atm. pressure	101 325 Pa	Standard conditions
Air density	1.225 kg/m^3	Standard dry air

Table 6: CFD conditions

The simulations will give us several types of information that are essential to improve the bodywork.

We will visualise how air pressure is distributed over the vehicle, showing where air builds up or separates. We will also see the air velocity zones around the vehicle to identify the vortices and the turbulent "wake" areas at the rear that create drag.

The software will compute the total forces acting on the car (drag, downforce) and give us coefficients that summarise its aerodynamic performance. We will also study the pressure distribution over all surfaces (front, sides, flat floor) and visualise the streamlines to understand exactly how air flows around the vehicle.

These results will let us pinpoint the areas to modify to reduce drag and increase downforce.

Phase 5 – Iterative optimisation

Based on the CFD results, we will improve the bodywork step by step:

Step 1 – Reduce drag: smooth the front and rear shapes so that air does not build up, reduce the surfaces that "brake" the air at the front, and adjust the rear diffuser angle to better accelerate the outgoing air.

Step 2 – Increase downforce: optimise the flat floor to create more low pressure underneath (a suction-cup effect), adjust the rear wing for more downforce, and add small front canards that create stabilising vortices.

Step 3 – Balance the forces: check that downforce is well distributed between the front and rear wheels (neither too much at the front nor at the rear) and adjust the geometry if needed to reach a perfect balance.

Each change will be tested again in simulation until we reach the best compromise between top speed and cornering stability.

3 TECHNICAL SPECIFICATIONS – EMBEDDED F1-STYLE STEERING WHEEL

3.1 General context and objectives

The steering wheel is the main interface between the driver and the car. Inspired by Formula 1 steering wheels, it is a multifunction control centre that brings together all the vehicle's essential functions.

For an electric vehicle without a mechanical gearbox, the main challenge is to create the impression of a real transmission by simulating in software the feel of a conventional gearbox (progressive acceleration, speed plateaus per gear).

The software used for the mechatronics part is KiCad, Arduino IDE and Thonny.

3.2 Essential steering-wheel functions

Function 1: Steering the vehicle
Need: allow the driver to steer the vehicle and to get in and out of the cockpit easily thanks to a removable steering wheel, as on Formula 1 cars.

Description: the steering wheel must be mounted on a quick-release hub. The driver must be able to remove the wheel to get in or out of the car, then refit it within a few seconds before driving. Once locked, the system must provide the same steering precision as a conventional connection.

Criteria:

- "Quick-release" system (pull / push / turn).
- Full fitting or removal time: a few seconds, achievable with gloves.
- Secure mechanical locking: the car cannot be driven if the wheel is not properly engaged.
- No perceptible play in the connection once locked (precise steering).

Function 2: Simulated gear changes
Need: simulate a virtual 6-speed gearbox.

Description:

- Right paddle: shift up (e.g. 3rd → 4th)
- Left paddle: shift down
- Automatic check of the allowed speed for each gear

Criteria:

- Paddle response: < 50 ms
- Speed plateaus:
  - G1: 50 km/h max
  - G2: 85 km/h max

- G3: 120 km/h max
- G4: 155 km/h max
- G5: 190 km/h max
- G6: 230 km/h max

Function 3: Powertrain strategies
Need: change the operating modes of the drivetrain.

Description: the steering wheel must let the driver easily select the vehicle's main driving mode and, when "Drive" is active, the associated power-management sub-mode.

Criteria:

- Mandatory main modes:
  - **D**: Drive (forward)
  - **R**: Reverse
  - **N**: Neutral
  - **P**: Park (immobilised)
- In D mode, two sub-modes must be available:
  - **Automatic mode:** standard power management for normal driving.
  - **Simulation mode:** activates the virtual gearbox logic (simulated gears, sporty feel).
- Fast switching between modes and sub-modes.
- Clear, permanent display on the steering-wheel screen of the main mode (D/R/N/P) and of the active sub-mode (Auto or Simulation).

Function 4: Radio communication
Need: communicate with the technical team.

Criteria:

- Dedicated radio button
- "Mute" button to cut the audio

Function 5: Accelerator pedal acquisition
Need: measure the pedal position precisely for motor control.

Criteria:

- Redundant Hall-effect sensor (V1 + V2)
- Accuracy: $\pm 0.5\,\%$
- Fault detection: V1/V2 divergence $> 7\,\%$
- Safety: acceleration cut if a fault is detected

Function 6: Real-time information display
Need: give the driver all essential data.
Criteria:

- 3.5-inch TFT screen (480×320 px) at the centre of the wheel showing:
  - Vehicle speed (km/h)
  - Virtual engine speed (rpm)
  - Active driving mode
  - Battery charge (%)
  - System alerts
- Second screen connected to the motor driver to display the engaged gear

3.3 Control layout

Control	Position
Up/Down shift paddles	Back of the wheel
Right indicator	Top right
Left indicator	Top left
Radio/Mute button	Centre
Headlights on/off	Middle left
Headlight flash	Middle right
Powertrain strategies	Top of the wheel

Table 7: Complete layout

3.4 Safety criteria (essential)

Test	Acceptance criterion
Paddle response	< 50 ms
Pedal accuracy	$\pm 0.5\,\%$
Sensor redundancy	V1/V2 always active
Fault detection	< 20 ms
Button lifetime	10 000 presses
Electronics MTBF	50 000 hours

Table 8: Validation criteria

All these functions are essential and must be delivered to validate the steering-wheel project.